

Benchao Zhu

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RESEARCH INTEREST

Electronic Design Automation

EDUCATION

Fudan University

M.E. in Electronics and Information Engineering

Shanghai, CHN

September 2022 - June 2025

Fudan University

B.E. in Microelectronics Science and Engineering

Shanghai, CHN

September 2018 - June 2022

HONORS & AWARDS

Contest Awards

ISPD 2023 Contest (Security Closure of Physical Layouts)	First Place
AICAS 2025 Grand Challenge (LLM-Based Analog Circuit Design)	Third Place
ICCAD 2024 Contest (Power and Timing Optimization Using Multibit Flip-Flop)	Fourth Place
ISPD 2024 Contest (GPU/ML-Enhanced Large Scale Global Routing)	Fifth Place

Scholarships

Xiaomi Special Scholarship, Fudan University (10 recipients university-wide)	2023
National Encouragement Scholarship	2019–2021

PUBLICATIONS

- **An Analytical Placement Framework for Advanced Hybrid-Row-Height Circuit Designs**
Benchao Zhu, Yuan Wen, Jiarui Chen, Jianli Chen, and Zhifeng Lin
ACM Transactions on Design Automation of Electronic Systems (TODAES), 2026.
- **A Physically-aware Framework for Joint MBFF Synthesis with OPTICS-based Debanking**
Benchao Zhu, Jiawei Li, Yang Liu, Jianli Chen, and Keren Zhu
ACM Great Lakes Symposium on VLSI (GLSVLSI), 2026.
- **Layer-Aware Timing-Driven Global Routing for Advanced Technology Nodes**
Zhaoyi Wu, Haishan Huang, Benchao Zhu, Jianli Chen, and Zhifeng Lin
IEEE International Symposium on Circuits and Systems (ISCAS), 2026.
- **Legalization Framework with Design Rule Constraints Enhanced by Monte-Carlo-based Cell Priority Optimization**
Benchao Zhu, Guohao Chen, Jiawei Li, Peng Zou, Haokai Sun, Zhifeng Lin, Jianli Chen
IEEE International Symposium on Circuits and Systems (ISCAS), 2025.
- **Late Breaking Results: Mixed-Cell-Height Detailed Placement under Multi-Cell Spacing Constraints**
Benchao Zhu, Zheng Zeng, and Jianli Chen
ACM/IEEE Design Automation Conference (DAC), 2024.
- **Effective Analytical Placement for Advanced Hybrid-Row-Height Circuit Designs**
Yuan Wen, Benchao Zhu, Zhifeng Lin, and Jianli Chen
Asia and South Pacific Design Automation Conference (ASP-DAC), 2024.
- **Mixed-cell-height Placement with Minimum-Implant-Area and Drain-to-Drain Abutment Constraints**
Guohao Chen, Zheng Zeng, Benchao Zhu, Jiawei Li, Kun Wang, Jun Yu, and Jianli Chen
ACM/IEEE Design Automation Conference (DAC), 2023.